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Pengembangan Web Praktik

DATABASE MIGRATION





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Materi Perkuliahan

1

2

3 Dasar UI/UX Design, Instalasi Flask

4 Integrasi, HTML & CSS Dasar

5 Integrasi Flask & MySQL CRUD

6 Penerapan Session pada Flask

7 Responsi

8 Arsitektur MVC pada Flask

9 Konsep microservices dan Rest API pada Flask

10 Migration Model CRUD Rest API pada Flask

11 Penerapan API JWT pada Flask

12 Integrasi Flask dan Javascript

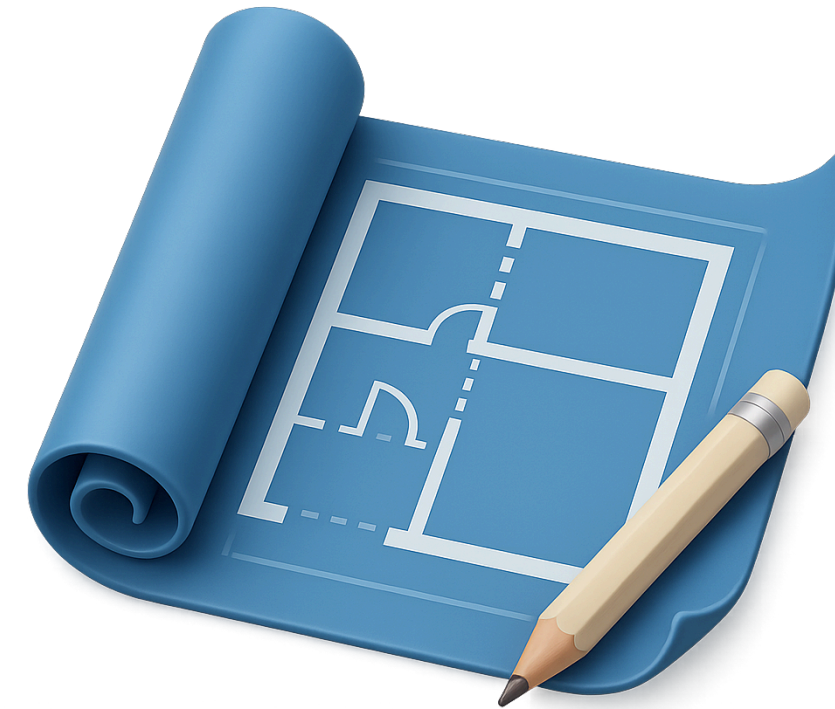
13 Presentasi Proyek Besar

14 Presentasi Proyek Besar



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SQLAlchemy

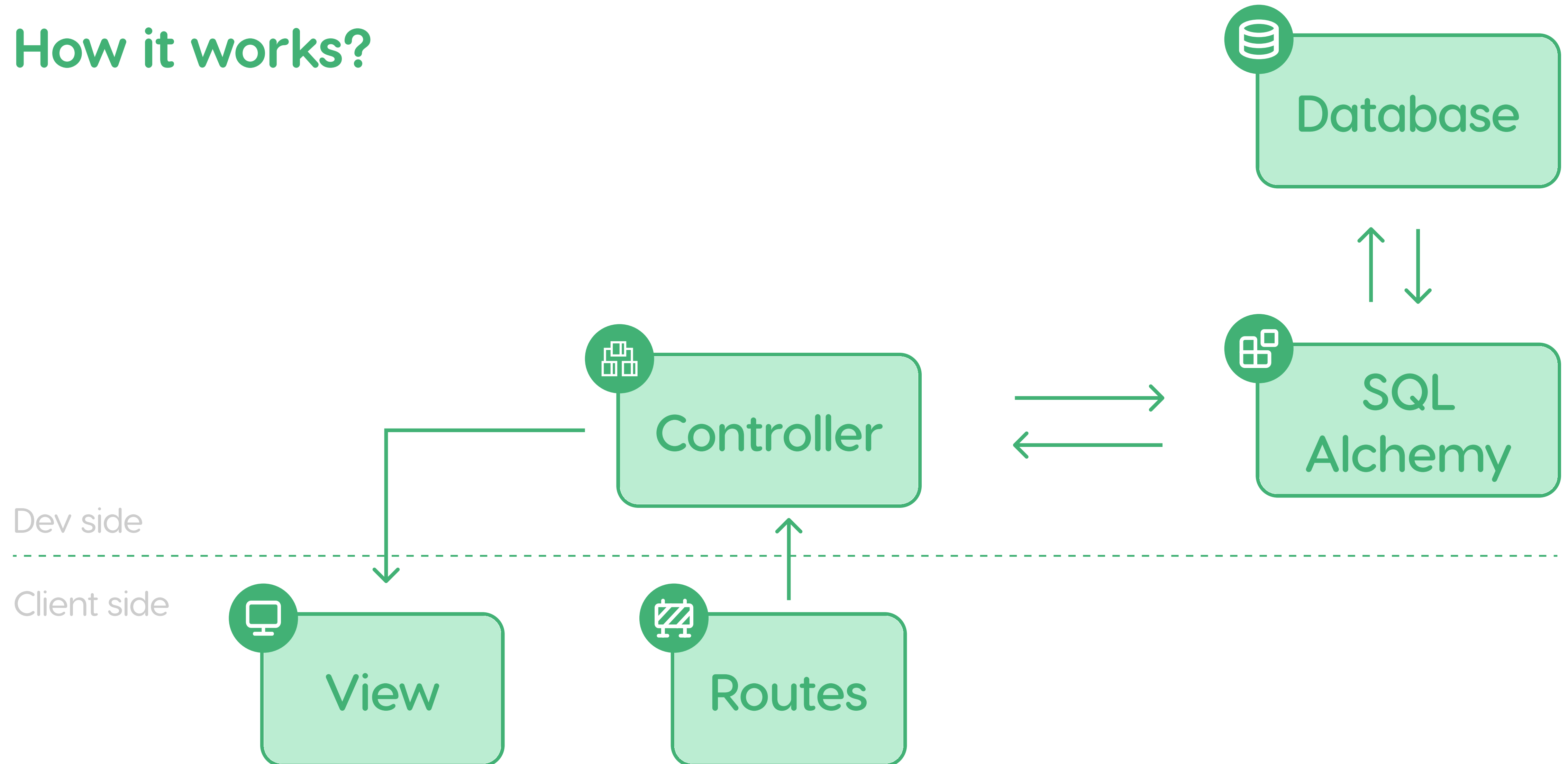


SQLAlchemy provides an **Object Relational Mapper (ORM)** to interact with relational databases using Python classes, simplifying database operations. It also offers for writing SQL queries in Python and **supports various databases, schema management, and connection pooling.**



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How it works?





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SQLAlchemy analogy in building home

Documentation of renovation

Migration versioning

Layout changes

Changing table structure

Moving things around

Data Migration

Additional room space

Add new column



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How to Implement

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Installation

Flask sql mapping module

\$ pip install flask-migrate flask-sqlalchemy

Flask migration module

Configuration

A code editor window with a light gray background and a dark gray border. It has three colored window control buttons (red, yellow, green) in the top left corner. The code is written in a monospaced font with syntax highlighting: keywords like 'Flask', 'False', and 'True' are in blue; strings are in green; and the code structure is indented. Line numbers 1 through 11 are on the left side of the code block.

```
1  app = Flask(__name__)
2  app.config["MYSQL_HOST"] = "localhost"
3  app.config["MYSQL_PORT"] = "3306"
4  app.config["MYSQL_USER"] = "root"
5  app.config["MYSQL_PASSWORD"] = ""
6  app.config["MYSQL_DB"] = "db_books"
7  app.config['SQLALCHEMY_DATABASE_URI'] = f'mysql+pymysql://root:@localhost:3306/db_books'
8  app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
9  app.config['SQLALCHEMY_RECORD_QUERIES'] = True
10
11
```

Create Model

```
1  from myproject.app import del_book
2
3
4  class Books(db.Model):
5      __tablename__ = 'books'
6      id = db.Column(db.Integer, primary_key=True)
7      title = db.Column(db.String(50), nullable=False)
8      author = db.Column(db.String(30), nullable=False)
9      created_at = db.Column(db.DateTime, server_default=db.func.now())
10
11     def __init__(self, title, author):
12         self.title = title
13         self.author = author
14
15     def __repr__(self):
16         return f'<Book {self.title}>'
```



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Database Initialization

```
$ flask db init
```

Database Schema Changes

```
$ flask db migrate -m "some changes"
```



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Upgrade Schema

```
$ flask db upgrade
```

Downgrade Schema

```
$ flask db downgrade
```



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Let's dive into Flask!

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